
Pedro Henrique Bugatti, Ph.D.

Associate Professor

Department of Computing

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Research Summary

Researcher with extensive experience in Artificial Intelligence, Machine Learning, and Computer Vision. My main focus lies in developing perceptually aligned and data-efficient deep learning models for complex data domains such as biomedical images, biological sequences, and agricultural monitoring. With over 15 years of teaching and research experience, I have coordinated graduate programs, supervised MSc and PhD students, and led projects funded by national research agencies. I aim to advance AI-driven perception and knowledge extraction for interdisciplinary applications.

Education

- Postdoctoral Research (Visiting Scholar), Institute for Advanced Computer Studies (UMI-ACS), University of Maryland (USA), 2018.
- Ph.D. in Computer Science, University of São Paulo (ICMC-USP), Brazil, 2012. Advisor: Agma Juci Machado Traina.
- M.Sc. in Computer Science, University of São Paulo (ICMC-USP), Brazil, 2008. Advisor: Agma Juci Machado Traina.
- B.Sc. in Computer Science, University of Marília (UNIVEM), Brazil, 2005.

Academic Positions

- Associate Professor, Department of Computing, Federal University of São Carlos (UFSCar), Dec 2022–Present.
- Associate Professor, Federal University of Technology – Paraná (UTFPR), Dec 2010–Nov 2022.
- Deputy Coordinator, Graduate Program in Bioinformatics, UTFPR, Feb 2020–2022.
- Deputy Coordinator, Graduate Program in Computer Science, UTFPR, Feb 2016–Dec 2017.

Research Interests

- Artificial Intelligence, Machine Learning, Deep Learning, and Graph Neural Networks.
- Computer Vision and Image/Video Analysis.
- Data Mining and Pattern Recognition for Complex Data.

Teaching Experience

- Courses taught: Machine Learning, Image Processing, Computer Vision, Data Mining, Database Systems, Software Architecture, Algorithms, and Programming.
- Experience teaching at undergraduate and graduate levels.

Academic Service

- Reviewer for major journals: *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *IEEE Transactions on Knowledge and Data Engineering*, *IEEE Transactions on Industrial Electronics*, *IEEE Transactions on Industrial Informatics*, *IEEE Transactions on Emerging Topics in Computational Intelligence*, *IEEE Journal of Biomedical and Health Informatics*, *Pattern Recognition Letters*, *Journal of Medical Imaging*, *Briefings in Bioinformatics*, *Artificial Intelligence in Medicine*, *Plos One*, *Computers in Biology and Medicine*, *Computers and Electronics in Agriculture*, *Revista de Informática Teórica e Aplicada (RITA)*, *Expert Systems with Applications*, *Applied Soft Computing*, *Computational and Structural Biotechnology Journal*, *Artificial Intelligence Review*, *Journal of Ambient Intelligence and Humanized Computing*, *Biosystems Engineering*, *Image and Vision Computing*, *Computer Methods and Programs in Biomedicine*, *IEEE Access*, *Entropy*, *Applied Artificial Intelligence*, *Symmetry*, *PeerJ Computer Science*, *Innovation and Research in Biomedical Engineering*, *Applied Sciences*, among others.
- Program Committee Member: Program Committee Member: Conference on Graphics, Patterns and Images (SIBGRAPI), Iberoamerican Congress on Pattern Recognition (CIARP), International Conference on Computer Vision Theory and Applications (VISAPP), IADIS International Conference on Applied Computing, and Brazilian e-Science Workshop.
- Advisory Committees: National Council for Scientific and Technological Development (CNPq), São Paulo Research Foundation (FAPESP), Araucária Foundation of Support to the Scientific and Technological Development of Paraná.

Honors and Awards

- First Place in the XV Workshop on Computer Vision Thesis and Dissertations Contest (MSc. Student Rafael S. Bressan) (2019)
- Finalist of the 34th Brazilian Symposium on Databases Theses and Dissertations Contest, Brazilian Symposium on Databases (MSc. Student Rafael S. Bressan) (2019)
- Best Paper Award - IX Demos and Applications Session of the 27th Brazilian Symposium on Databases (SBBD), 2012.
- Among the 10 best MSc. dissertations in the country, Brazilian Computer Society (SBC), 2008.
- Outstanding Student Award, Brazilian Computer Society (SBC), 2005.
- Best Student Award - BSc. in Computer Science (2006).

Journal Publications

1. A. Gaspar Chiquitto, L. S. Oliveira, **P. H. Bugatti**, P. T. M. Saito, M. Basham, R. T. Raittz, A. R. Paschoal. *GENNUS: Generative Approaches for Nucleotide Sequences*

- Enhance Mirtron Classification*. **NAR Genomics and Bioinformatics**, vol. 7, pp. 1–12 (lqaf072), 2025.
2. S. R. R. Sanches, E. Custódio Jr., C. G. Corrêa, C. Oliveira, V. Freire, P. T. M. Saito, **P. H. Bugatti**. *Automatic Generation of Difficulty Maps for Datasets Using Neural Networks*. **Multimedia Tools and Applications**, pp. 1–18, 2024.
 3. S. R. R. Sanches, C. G. Corrêa, B. R. Brum, **P. H. Bugatti**, P. T. M. Saito, C. M. Silva, E. Custódio Jr. *Evaluation of Change Detection Algorithms Using Difficulty Maps*. **IEEE Latin America Transactions**, vol. 21, pp. 700–706, 2023.
 4. E. R. Oliveira, **P. H. Bugatti**, P. T. M. Saito. *Assessment of Clustering Techniques to Support the Analyses of Soybean Seed Vigor*. **PLoS One**, vol. 18, pp. 1–20, 2023.
 5. M. Souza Jr., W. C. Horikoshi, P. T. M. Saito, **P. H. Bugatti**. *Soybean Seed Vigor Classification Through an Effective Image Learning-Based Approach*. **Multimedia Tools and Applications**, pp. 1–12, 2023.
 6. R. S. Moura, S. R. R. Sanches, **P. H. Bugatti**, P. T. M. Saito. *Pedestrian Traffic Lights and Crosswalk Identification*. **Multimedia Tools and Applications**, pp. 1–10, 2022.
 7. R. P. P. de Souza, C. M. Agulhari, A. Goedtel, **P. H. Bugatti**. *Improved Robust Filter Synthesis Combining Linear Matrix Inequalities and Heuristic Algorithms*. **Optimal Control Applications and Methods**, pp. 1–16, 2022.
 8. L. G. Batista, **P. H. Bugatti**, P. T. M. Saito. *Classification of Skin Lesions Through Active Learning Strategies*. **Computer Methods and Programs in Biomedicine**, vol. 226, 107122, 2022.
 9. C. M. Silva, K. A. I. Rosa, **P. H. Bugatti**, P. T. M. Saito, C. G. Corrêa, R. S. Yokoyama, S. R. R. Sanches. *Method for Selecting Representative Videos for Change Detection Datasets*. **Multimedia Tools and Applications**, vol. 81, pp. 3773–3791, 2022.
 10. D. F. Pereira, **P. H. Bugatti**, F. M. Lopes, A. L. S. M. de Souza, P. T. M. Saito. *Assessing Active Learning Strategies to Improve the Quality Control of the Soybean Seed Vigor*. **IEEE Transactions on Industrial Electronics**, vol. 68, pp. 1675–1683, 2021.
 11. G. Camargo, **P. H. Bugatti**, P. T. M. Saito. *Active Semi-Supervised Learning for Biological Data Classification*. **PLoS One**, vol. 15, e0237428, 2020.
 12. M. H. P. da Cruz, D. S. Domingues, P. T. M. Saito, A. R. Paschoal, **P. H. Bugatti**. *TERL: Classification of Transposable Elements by Convolutional Neural Networks*. **Briefings in Bioinformatics**, vol. 22, bbaa185, 2020.
 13. D. F. Pereira, **P. H. Bugatti**, F. M. Lopes, A. L. S. M. Souza, P. T. M. Saito. *Contributing to Agriculture by Using Soybean Seed Data from the Tetrazolium Test*. **Data in Brief**, vol. 23, 103652, 2019.
 14. G. V. L. de Lima, P. T. M. Saito, F. M. Lopes, **P. H. Bugatti**. *Classification of Texture Based on Bag-of-Visual-Words Through Complex Networks*. **Expert Systems with Applications**, vol. 133, pp. 215–224, 2019.
 15. R. S. Bressan, **P. H. Bugatti**, P. T. M. Saito. *Breast Cancer Diagnosis Through Active Learning in Content-Based Image Retrieval*. **Neurocomputing**, vol. 357, pp. 1–10, 2019.
 16. R. S. Bressan, G. Camargo, **P. H. Bugatti**, P. T. M. Saito. *Exploring Active Learning Based on Representativeness and Uncertainty for Biomedical Data Classification*. **IEEE Journal of Biomedical and Health Informatics**, vol. 23, pp. 2238–2244, 2019.

17. T. C. Negri, W. A. L. Alves, **P. H. Bugatti**, P. T. M. Saito, D. S. Domingues, A. R. Paschoal. *Pattern Recognition Analysis on Long Noncoding RNAs: A Tool for Prediction in Plants*. **Briefings in Bioinformatics**, vol. 20, pp. 682–689, 2019.
18. **P. H. Bugatti**, D. S. Kaster, M. P. Silva, A. J. M. Traina, C. Traina Jr. *PRoSPer: Perceptual Similarity Queries in Medical CBIR Systems Through User Profiles*. **Computers in Biology and Medicine**, vol. 45, pp. 8–19, 2014.
19. D. S. Kaster, **P. H. Bugatti**, A. J. M. Traina, C. Traina Jr. *FMI-SiR: A Flexible and Efficient Module for Similarity Searching on Oracle Database*. **Journal of Information and Data Management (JIDM)**, vol. 1, pp. 229–244, 2010.
20. M. X. Ribeiro, **P. H. Bugatti**, C. Traina Jr., P. M. A. Marques, N. A. Rosa, A. J. M. Traina. *Supporting Content-Based Image Retrieval and Computer-Aided Diagnosis Systems with Association Rule-Based Techniques*. **Data & Knowledge Engineering**, vol. 68, pp. 1370–1382, 2009.

Conference Publications

1. M. M. Fernandez, A. L. M. Levada, **P. H. Bugatti**. *Deep Learning for Protoplanetary Disk Classification from Hubble Images: A Comparative Study on Imbalanced Data*. In: **38th Conference on Graphics, Patterns and Images (SIBGRAPI)**, 2025, Salvador, Brazil, pp. 1-6.
2. R. C. Masson, K. R. L. J. C. Branco, **P. H. Bugatti**, P. T. M. Saito, R. A. S. Arruda. *Analysis of Data and Supervised Machine Learning Algorithms for Intrusion Detection*. In: **24th International Conference on Artificial Intelligence and Soft Computing (ICAISC)**, 2025, Poland, Springer, pp. 1–8.
3. L. G. Souza, S. R. R. Sanches, **P. H. Bugatti**, P. T. M. Saito. *Combining Audio and Image Sequence for Video Moment Retrieval by Natural Language*. In: **23rd International Conference on Artificial Intelligence and Soft Computing (ICAISC)**, 2024, Zakopane, Poland. Springer, pp. 1–12.
4. A. L. Franco, C. M. Agulhari, **P. H. Bugatti**, P. T. M. Saito. *Classification of Cardiac Arrhythmias Using 1D and 2D Convolutional Neural Networks*. In: **23rd ICAISC**, 2024, Zakopane, Poland. Springer, pp. 1–11.
5. W. S. M. Silva, P. T. M. Saito, **P. H. Bugatti**. *Image Classification Through Graph Neural Networks and Random Walks*. In: **22nd ICAISC**, 2023, Zakopane, Poland, pp. 1–10.
6. C. N. O. Bassani, P. T. M. Saito, **P. H. Bugatti**. *Assessment of Semi-Supervised Approaches Applied to Convolutional Neural Networks*. In: **21st ICAISC**, 2022, Zakopane, Poland, pp. 195–205.
7. L. G. Batista, F. F. Cruz, **P. H. Bugatti**, P. T. M. Saito. *Aprendizado Ativo Aplicado ao Diagnóstico de Lesões Dermatológicas*. In: **XVI Brazilian e-Science Workshop (BreSci)**, 2022, Rio de Janeiro, pp. 1–8.
8. C. N. O. Bassani, P. T. M. Saito, **P. H. Bugatti**. *Avaliação de Abordagens Semi-Supervisionadas Aplicadas a Redes Neurais Convolucionais*. In: **XVI BreSci**, 2022, Rio de Janeiro, pp. 9–16.
9. F. F. Costa, P. T. M. Saito, **P. H. Bugatti**. *Video Action Classification Through Graph Convolutional Networks*. In: **16th International Conference on Computer Vision**

Theory and Applications (VISAPP), 2021, pp. 490–497.

10. J. M. Tozato, **P. H. Bugatti**, P. T. M. Saito. *Active Learning Strategies and Convolutional Neural Networks for Mammogram Classification*. In: **20th ICAISC**, 2021, Zakopane, Poland. Springer LNCS 12855, pp. 126–134.
11. W. S. Campos, L. G. Souza, P. T. M. Saito, **P. H. Bugatti**. *Contextual Image Classification Through Fine-Tuned Graph Neural Networks*. In: **20th ICAISC**, 2021, Zakopane, Poland. Springer LNCS 12855, pp. 15–24.
12. M. S. Silva, L. F. Cruz, **P. H. Bugatti**, P. T. M. Saito. *Automatic Visual Quality Assessment of Biscuits Using Machine Learning*. In: **19th ICAISC**, 2020, Zakopane, Poland. Springer LNAI, pp. 59–70.
13. L. F. Cruz, P. T. M. Saito, **P. H. Bugatti**. *DeepCloud: An Investigation of Geostationary Satellite Imagery Frame Interpolation for Improved Temporal Resolution*. In: **19th ICAISC**, 2020, Zakopane, Poland. Springer LNCS, pp. 50–59.
14. D. F. Pereira, **P. H. Bugatti**, F. M. Lopes, A. L. S. M. Souza, P. T. M. Saito. *An Intelligent System to Enhance the Productivity and Sustainability in Soybean Crop Enterprises*. In: **XV Brazilian Symposium on Information Systems (SBSI)**, 2019, Aracaju, pp. 1–8.
15. L. M. Valerio, D. H. A. Alves, L. F. Cruz, **P. H. Bugatti**, C. Oliveira, P. T. M. Saito. *DeepMammo: Deep Transfer Learning for Lesion Classification of Mammographic Images*. In: **32nd IEEE International Symposium on Computer-Based Medical Systems (CBMS)**, 2019, Córdoba, Spain, pp. 447–452.
16. M. H. P. da Cruz, P. T. M. Saito, A. R. Paschoal, **P. H. Bugatti**. *Classification of Transposable Elements by Convolutional Neural Networks*. In: **18th ICAISC**, 2019, Zakopane, Poland. Springer LNCS, pp. 157–168.
17. L. F. Cruz, P. T. M. Saito, **P. H. Bugatti**. *DeepSeed: A Deep Learning Methodology for Automated Soybean Seed Damage Classification*. In: **15th International Conference on Machine Learning and Data Mining (MLDM)**, 2019, New York, USA. Springer LNCS, pp. 890–900.
18. L. O. Alves, L. F. Cruz, P. T. M. Saito, **P. H. Bugatti**. *Towards Practical Computer Vision in Teaching and Learning of Image Processing Theories*. In: **IEEE Frontiers in Education**, 2019, Cincinnati, USA, pp. 1–7.
19. R. S. Bressan, **P. H. Bugatti**, P. T. M. Saito. *Abordagens de Aprendizado Ativo para Recuperação e Classificação de Imagens*. In: **34º SBBD**, 2019, Fortaleza, pp. 1–6.
20. D. H. A. Alves, L. F. Galonetti, C. Oliveira, **P. H. Bugatti**, P. T. M. Saito. *Going Deeper on BioImages Classification: A Plant Leaf Dataset Case Study*. In: **22nd Iberoamerican Congress on Pattern Recognition (CIARP)**, 2018, Valparaíso, Chile, pp. 36–44.
21. R. S. Bressan, D. H. A. Alves, L. M. Valerio, **P. H. Bugatti**, P. T. M. Saito. *DOCToR: The Role of Deep Features in Content-Based Mammographic Image Retrieval*. In: **31st IEEE CBMS**, 2018, Karlstad, Sweden, pp. 158–163.
22. M. Souza Jr., R. S. Bressan, D. F. Pereira, P. T. M. Saito, **P. H. Bugatti**. *Rumo à Melhoria de Produtividade e Sustentabilidade Agrícola por Meio da Classificação Automática do Vigor de Sementes de Soja*. In: **45º SEMISH**, 2018, Natal, pp. 113–124.
23. R. S. Bressan, M. Souza Jr., D. F. Pereira, **P. H. Bugatti**, P. T. M. Saito. *SoyRetrieval – Técnicas de Aprendizado e Recuperação de Imagens para Análise do Vigor de Sementes*

- de Soja. In: **45^o SEMISH**, 2018, Natal, pp. 77–88.
24. G. Camargo, R. S. Bressan, **P. H. Bugatti**, P. T. M. Saito. *Towards an Effective and Efficient Learning for Biomedical Data Classification*. In: **30th IEEE CBMS**, 2017, Thessaloniki, Greece, pp. 13–18.
 25. D. F. Pereira, P. T. M. Saito, **P. H. Bugatti**. *An Image Analysis Framework for Effective Classification of Seed Damages*. In: **31st ACM Symposium on Applied Computing (SAC)**, 2016, Pisa, Italy, pp. 61–66.
 26. G. V. L. de Lima, T. R. Castilho, **P. H. Bugatti**, P. T. M. Saito, F. M. Lopes. *A Complex Network-Based Approach to the Analysis and Classification of Images*. In: **20th CIARP**, 2015, Montevideo, Uruguay. Springer LNCS 9423, pp. 322–330.
 27. R. Rocha, P. T. M. Saito, **P. H. Bugatti**. *A Novel Framework for Content-Based Image Retrieval Through Relevance Feedback Optimization*. In: **20th CIARP**, 2015, Montevideo, Uruguay. Springer LNCS 9423, pp. 281–289.
 28. M. G. F. Sant’Anna, P. T. M. Saito, **P. H. Bugatti**. *Content-Based Image Retrieval Towards the Automatic Characterization of Soybean Seed Vigor*. In: **29th ACM SAC**, 2014, Gyeongju, South Korea, pp. 964–969.
 29. M. A. P. Terrin, C. N. Silla Jr., **P. H. Bugatti**. *Utilizando Técnicas de Mineração de Dados Para Apoiar a Busca Ativa de Famílias em Situação de Vulnerabilidade e Risco Social*. In: **XLI SEMISH**, 2014, Brasília, pp. 1051–1062.
 30. P. H. Bugatti *et al.* *Integrating User Profile in Medical CBIR Systems to Answer Perceptual Similarity Queries*. In: **SPIE Medical Imaging**, 2011, Orlando, USA, pp. 1–12.

Book Chapters

1. R. S. Bressan, **P. H. Bugatti**, P. T. M. Saito. *Optimum-Path Forest and Active Learning Approaches for Content-Based Medical Image Retrieval*. In: A. X. Falcão, J. P. Papa (Eds.), **Optimum-Path Forest: Theory, Algorithms, and Applications**. Academic Press, 2022, pp. 95–207.
2. L. S. Oliveira, A. C. Patera, D. S. Domingues, D. S. Sanches, **P. H. Bugatti**, P. T. M. Saito, V. Maracajá-Coutinho, A. M. Durham, A. R. Paschoal. *Computational Analysis of Transposable Elements and circRNAs in Plants*. In: L. M. Vaschetto (Ed.), **Plant Circular RNAs: Methods and Protocols**. Springer US, New York, 2021, Vol. 2362, pp. 147–172.
3. A. J. M. Traina, C. Traina Jr., A. G. R. Balan, M. X. Ribeiro, **P. H. Bugatti**, C. Y. V. W. Silva, P. M. A. Marques. *Feature Extraction and Selection for Decision Making over Medical Images*. In: T. Deserno (Ed.), **Biomedical Image Processing – Methods and Applications**. Springer-Verlag, Berlin, 2011, pp. 181–209.
4. A. Talarico Neto, R. P. M. Fortes, F. C. Teixeira, L. C. D. Alves, **P. H. Bugatti**, O. A. Carvalho Jr. *Interfaces Multimodais Dinâmicas Utilizando J2EE*. In: M. R. S. Borges, J. G. P. Filho (Eds.), **Mini-Cursos WebMedia 2008**. Sociedade Brasileira de Computação (SBC), Vila Velha, ES, 2008, Vol. 1, pp. 1–48.

Research Grants

1. **Principal Investigator:** Simbio – Bioimages Similarity Queries Towards User’s Perception. CNPq Universal Grant, 2017–2020.

2. **Postdoctoral Project:** University of Maryland – UMIACS. University Grant, 2018.
3. **Principal Investigator:** Deep Learning Applied to Classification and Detection of Complex Data. University Grant. 2020–2023.
4. **Project Member:** Deep Learning Applied to Image Analysis. University Grant. 2017–2019.
5. **Project Member:** Bioinformatics – Gene Network Inference and Omics Sequence Recognition. CNPq Universal Grant. 2023–Present.
6. **Project Member:** TELearning. Grant STIC AmSud - 21-STIC-13, France and Latin American (Brazil, Chile and Colombia), 2021–Present.
7. **Project Member:** NAPI-NORTE – Artificial Intelligence Center for Agriculture (CIA-Agro), 2021–Present. **Grant:** Araucária Foundation.
8. **Project Member:** Learning Approaches for Generation and Classification of Complex Data, 2020–2023.
9. **Project Member:** ARP-BIO: Pattern Learning and Recognition in Bioimages, 2017–2020. CNPq Universal Grant.
10. **Project Member:** Processing, Visualization, and Analysis of Complex Data. University Grant, 2017–2020.
11. **Project Member:** Improving Agricultural Productivity through Bioimage Methods, Techniques, and Analysis (RHAЕ). CNPq RHAЕ Grant (Human Resources in Strategic Areas – Researcher in Industry), 2015–2017.

Supervision and Advising (Completed)

Supervision and co-supervision of students (MSc, PhD, and undergraduate) in AI-related topics, including machine learning, deep learning, graph-based modeling, computer vision and biomedical/agriculture image analysis.

Doctoral Thesis

1. Reginaldo da Rocha. *Classification of Soybean Seed Images with Multiple Labels Associated with Vigor*. Ph.D. in Bioinformatics (UFPR/UTFPR), Federal University of Technology – Paraná, 2025. Supervisor: **Pedro Henrique Bugatti**.

Master’s Dissertations

1. William Sdayle Marins Silva. *Context in Images through Random Walks and Graph Convolutional Networks*. M.Sc. in Bioinformatics, Federal University of Technology – Paraná, 2023. Supervisor: **Pedro Henrique Bugatti**.
2. Lucas Pereira Lira. *A Novel Image Descriptor Based on Cellular Automata*. M.Sc. in Computer Science, Federal University of Technology – Paraná, 2023. Co-supervisor: **Pedro Henrique Bugatti**.
3. Walacy da Silva Campos. *Contextual Image Classification Using Graph Convolutional Networks and Similarity Pruning*. M.Sc. in Computer Science, Federal University of Technology – Paraná, 2022. Supervisor: **Pedro Henrique Bugatti**.
4. Lucas Garcia Batista. *Analysis and Classification of Skin Lesions Using Active Learning Strategies*. M.Sc. in Computer Science, Federal University of Technology – Paraná, 2022. Co-supervisor: **Pedro Henrique Bugatti**.

5. Cristiano Neia de Oliveira Bassani. *Semi-Supervised Convolutional Neural Networks Applied to Domain Adaptation*. M.Sc. in Computer Science, Federal University of Technology – Paraná, 2022. Supervisor: **Pedro Henrique Bugatti**.
6. Elton Custodio Júnior. *Automatic Generation of Difficulty Maps for Datasets*. M.Sc. in Computer Science, Federal University of Technology – Paraná, 2022. Co-supervisor: **Pedro Henrique Bugatti**.
7. Felipe Franco Costa. *Video Action Classification Using Graph-Based Convolutional Neural Networks*. M.Sc. in Computer Science, Federal University of Technology – Paraná, 2020. Supervisor: **Pedro Henrique Bugatti**.
8. Murilo Horácio Pereira Cruz. *Classification of Transposable Elements Using Convolutional Neural Networks*. M.Sc. in Bioinformatics, Federal University of Technology – Paraná, 2020. Supervisor: **Pedro Henrique Bugatti**.
9. Lucas Messias Valério. *Deep Active Learning for Annotation and Diagnosis of Intestinal Parasitoses*. M.Sc. in Bioinformatics, Federal University of Technology – Paraná, 2020. Co-supervisor: **Pedro Henrique Bugatti**.
10. Marcelo de Souza Júnior. *Soybean Seed Image Classification Using Deep Learning*. M.Sc. in Computer Science, Federal University of Technology – Paraná, 2018. Supervisor: **Pedro Henrique Bugatti**.
11. Rafael Staiger Bressan. *Active Learning for Image Retrieval and Classification*. M.Sc. in Computer Science, Federal University of Technology – Paraná, 2018. Co-supervisor: **Pedro Henrique Bugatti**.
12. Guilherme Camargo. *Active Learning for Bioimage Classification*. M.Sc. in Bioinformatics, Federal University of Technology – Paraná, 2018. Co-supervisor: **Pedro Henrique Bugatti**.
13. Vinícius Picosse Teruel. *Uncovering Human Face Patterns for Avatar Representation*. M.Sc. in Computer Science, Federal University of Technology – Paraná, 2018. Co-supervisor: **Pedro Henrique Bugatti**.
14. Geovana Veloso Loureiro de Lima. *Bag-of-Visual Words through Complex Networks*. M.Sc. in Computer Science, Federal University of Technology – Paraná, 2017. Supervisor: **Pedro Henrique Bugatti**.
15. Reginaldo Rocha. *Exploring Evolutionary Approaches for Content-Based Image Retrieval*. M.Sc. in Computer Science, Federal University of Technology – Paraná, 2016. Supervisor: **Pedro Henrique Bugatti**.
16. Marcus Glaucio Faria de Sant’Anna. *Automatic Retrieval and Classification of Soybean Seed Vigor*. M.Sc. in Computer Science, Federal University of Technology – Paraná, 2015. Supervisor: **Pedro Henrique Bugatti**.
17. Marcos Alexandre Pastori Terrin. *Using Data Mining Techniques to Support the Active Search of Families in Social Vulnerability*. M.Sc. in Computer Science, Federal University of Technology – Paraná, 2015. Co-supervisor: **Pedro Henrique Bugatti**.

Undergraduate Research (Scientific Initiation)

Undergraduate research projects (some with grants from CNPq and Araucária Foundation)

1. Luigi Freitas Cruz. *Deep Learning Involving Graph Convolutional Networks*. Computer Engineering, UTFPR, CNPq, 2022. Supervisor: **Pedro Henrique Bugatti**.

2. Luigi Freitas Cruz. *Self-Supervised Learning in Remote Sensing Images*. Computer Engineering, UTFPR, CNPq, 2021. Supervisor: **Pedro Henrique Bugatti**.
3. Vitor Fabrile Guastala. *Deep Learning for Contextual Detection*. Computer Science, UTFPR, CNPq, 2020. Supervisor: **Pedro Henrique Bugatti**.
4. Felipe Seolin Bento. *Graph-Based Methods for Image Description*. Software Engineering, UTFPR, 2020. Supervisor: **Pedro Henrique Bugatti**.
5. Thalia Akemi Kojo. *Application of Meta-Learning Methods for Complex Data Retrieval*. Automation Engineering, UTFPR, 2020. Supervisor: **Pedro Henrique Bugatti**.
6. Luigi Freitas Cruz. *Exploring Convolutional Neural Networks for Soybean Seed Classification*. Computer Engineering, UTFPR, CNPq, 2019. Supervisor: **Pedro Henrique Bugatti**.
7. Murilo Horácio Pereira da Cruz. *Evolutionary Methods for Defining Deep Learning Architectures*. Computer Engineering, UTFPR, Araucária Foundation, 2017. Supervisor: **Pedro Henrique Bugatti**.
8. Yasmim Gouveia Martins. *Application of Latent Deep Learning*. Computer Engineering, UTFPR, Araucária Foundation, 2017. Supervisor: **Pedro Henrique Bugatti**.
9. Gustavo Viais de Brito Silva. *Analysis of Computer Vision Techniques*. Computer Engineering, UTFPR, Araucária Foundation, 2016. Supervisor: **Pedro Henrique Bugatti**.
10. Vitor Camargo Soares. *Image Classification on Resource-Constrained Devices*. Computer Engineering, UTFPR, 2016. Supervisor: **Pedro Henrique Bugatti**.
11. William Renan Debrino. *Analysis of Reordered Similarity Search*. Computer Engineering, UTFPR, 2016. Supervisor: **Pedro Henrique Bugatti**.
12. Gianluca Maziero Machado. *Graph-Based Image Description Techniques*. Computer Engineering, UTFPR, 2016. Supervisor: **Pedro Henrique Bugatti**.
13. Marcelo Ortiz Paglione Junior. *Study and Analysis of Image Segmentation Techniques*. Computer Engineering, UTFPR, 2016. Supervisor: **Pedro Henrique Bugatti**.
14. Willian Cardoso Horikoshi. *User-Oriented Image Queries*. Computer Engineering, UTFPR, 2015. Supervisor: **Pedro Henrique Bugatti**.
15. Renata Carina Soares. *Effective Approaches for Image Representation*. Computer Engineering, UTFPR, 2015. Supervisor: **Pedro Henrique Bugatti**.
16. Gustavo Viais de Brito Silva. *Study and Development of Similarity Queries*. Computer Engineering, UTFPR, 2014. Supervisor: **Pedro Henrique Bugatti**.
17. William Cardoso Horikoshi. *Efficient Methods for Image Retrieval*. Computer Engineering, UTFPR, 2014. Supervisor: **Pedro Henrique Bugatti**.
18. Renata Carina Soares. *Study and Analysis of Methods for Image Description*. Computer Engineering, UTFPR, 2014. Supervisor: **Pedro Henrique Bugatti**.
19. Douglas Felipe Pereira. *Processing and Retrieval of Complex Data Applied to Soybean Seeds*. Computer Engineering, UTFPR, Araucária Foundation, 2014. Supervisor: **Pedro Henrique Bugatti**.

Undergraduate Theses (Bachelor's Projects)

1. Thalia Akemi Kojo. *Remaining Useful Life Prediction in Lithium-Ion Batteries Using Metalearning Methodology*. B.Eng. in Automation Engineering, UTFPR, 2021. Supervisor: **Pedro Henrique Bugatti**.
2. Marília Machado Fernandez. *Proposal of an Image Descriptor Based on Hilbert Curves*. B.Eng. in Computer Engineering, UTFPR, 2021. Supervisor: **Pedro Henrique Bugatti**.
3. Felipe Bento Seolin. *Texture Description Model for Images Based on Minimum Spanning Trees*. B.Eng. in Software Engineering, UTFPR, 2021. Supervisor: **Pedro Henrique Bugatti**.
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Additional Information

Comprehensive records of publications, research supervision, research funding, professional activities, among other details, are available at: **Lattes CV (CNPq)**